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Date Submitted: 11/ 30 Time Submitted 9:00

CSE 3341 – Digital Logic Design II
CSE 5357 – Advanced Digital Logic Design

Fall Semester 2025

Term Project

Half-Precision Floating-Point Adder Subtractor

(100+ Points)

Due by December 2, 2025, 11:59 pm

Submit through Canvas Assignments

Submission Requirements

- Submit the entire Quartus project in a zipped folder to ensure all files are included for execution.
- Name the zip file as `netid\_assignment\_x` where `netid` is your university network ID, and `x` corresponds to the assignment number (e.g., `jdx1234\_assignment\_1`).
- Make sure the project files can be run seamlessly on another machine without any missing dependencies.

Contact the TA for any clarification or help.

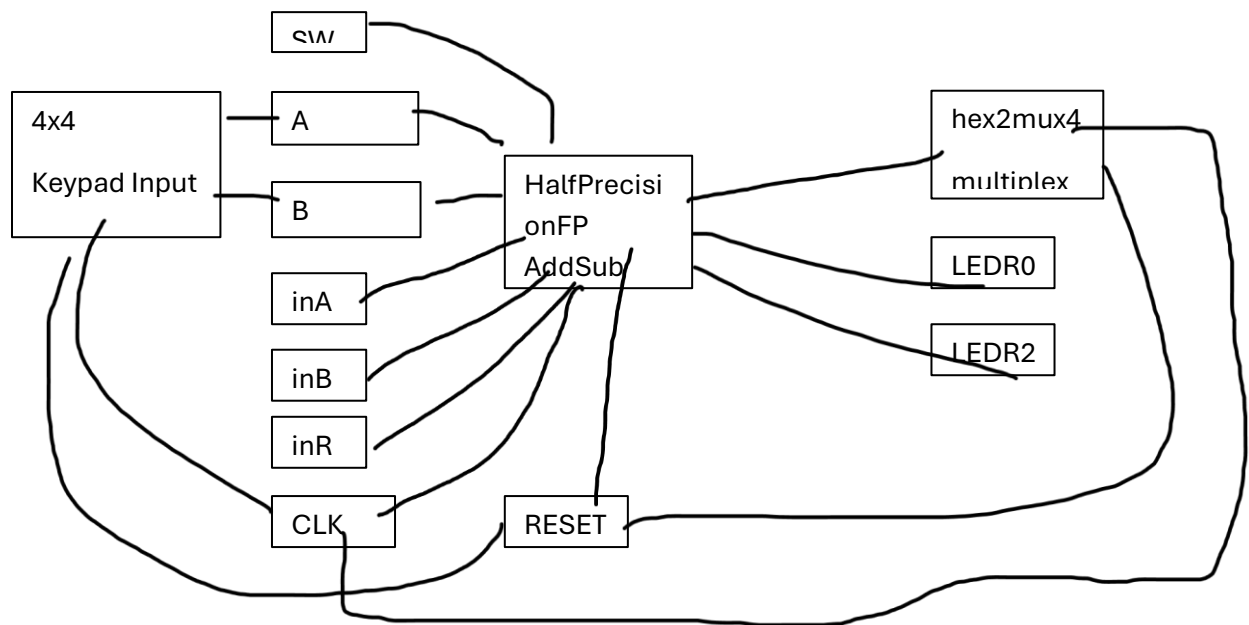
2. Summary & Project Description

This project implements an IEEE 754 Half-Precision Float-Point Adder/Subtractor on DE10-FPGA. The system takes two 16bit inputs (OpA, OpB) in hex and display the result in hex. Switch debounced.

2.1 Implemented Features

- Adder/Subtractor
 - > **SW0** to toggle operation. Add(0), Sub(1).
 - > Mantissa Norm to handle leading 0's and adjust exponent accordingly.
- Overflow/ Underflow
 - > **LEDR0** for Underflow, triggers when exponent become 00000 after Norm.
 - Or when the difference btw exponents is larger than mantissa bits.
 - > **LEDR2** for Overflow, triggers when the exponent becomes 11111, infinity.

3. Block diagram



4. DEMONSTRATION

TEST INPUT

$$\begin{aligned} 3.75(10) &= 11.11 (2) \\ &= 1.111 * 2^1 (2) \\ &= 0\ 10000\ 111000\ 0000 \\ &= 0x4380 \end{aligned}$$

$$\begin{aligned} 5.125(10) &= 101.001 (2) \\ &= 1.01001 * 2^2 (2) \\ &= 0\ 10001\ 010010\ 0000 \\ &= 0x4520 \end{aligned}$$

$$\begin{aligned} 9.5(10) &= 1001.1 (2) \\ &= 1.0-11 * 2^3 \\ &= 0\ 10010\ 001100\ 0000 \\ &= 0x48C0 \end{aligned}$$

$$\begin{aligned} 0.5625(10) &= 0.1001(2) \\ &= 1.001 * 2^{-1} \\ &= 0\ 01110\ 001000\ 0000 \\ &= 0x3880 \end{aligned}$$

$$\begin{aligned} 32768(10) &= 1.0 * 2^{15} (2) \\ &= 0\ 11110\ 0000000000 \\ &= 0x7800 \end{aligned}$$

$$\begin{aligned} 8.875(10) &= 1000.111(2) \\ &= 1.000111 * 2^3 \\ &= 0\ 10010\ 0001110000 \\ &= 0x4870 \end{aligned}$$

$$\begin{aligned} -1.375(10) &= -1.011(2) \\ &= 1\ 01111\ 0110000\ 000 \\ &= 0xBD80 \end{aligned}$$

$$10.0625(10) = 1010.0001(2)$$

$$\begin{aligned} 125.125(10) &= 1111101.001 (2) \\ &= 1.111101001 * 2^6 \\ &= 0\ 10101\ 111101001\ 0 \\ &= 0x57D2 \end{aligned}$$

$$\begin{aligned} 12.0625(10) &= 1100.0001 (2) \\ &= 1.1000001 * 2^3 \\ &= 0\ 10010\ 1000001\ 000 \\ &= 0x5A08 \end{aligned}$$

$$\begin{aligned} 1000(10) &= 1111\ 1010\ 00 (2) \\ &= 1.111101000 * 2^9 \\ &= 0\ 11000\ 1111010000 \\ &= 0x63D0 \end{aligned}$$

$$\begin{aligned} 1049(10) &= 10000011001 (2) \\ &= 1.0000011001 * 2^{10} \\ &= 0\ 11001\ 0000011001 \\ &= 0x6419 \end{aligned}$$

$$\begin{aligned} 0.000488758(10) &= 1.0 * 2^{-11} (2) \\ &= 0\ 00100\ 0000000001 \\ &= 0x0401 \end{aligned}$$

$$\begin{aligned} 8.9375(10) &= 1000.1111(2) \\ &= 1.0001111 * 2^3 \\ &= 0\ 100010\ 0001111000 \\ &= 0x4878 \end{aligned}$$

$$\begin{aligned} 113.0625(10) &= 1110001.0001(2) \\ &= 0\ 10101\ 1100010001 \\ &= 0x5711 \end{aligned}$$

$$2048(10) = 1.0 * 2^{11} (2)$$

= 1.0100001×2^3
 = 0 10010 010000 1000
 = 0x4908

= 0 11010 0000000000
 = 0x6800

OpA (Hex)	OpB (Hex)	operator	Result (Hex)	Result (Decimal)	overflow	underflow
0x4380	0x4520	0 / +	0x4870	8.875	x	x
		1 / -	0xBD80	-1.375	x	x
0x48C0	0x3880	0 / +	0x4908	10.0625	x	x
		1 / -	0x4878	8.9375	x	x
0x57D2	0x4A08	1 / -	0x5711	113.0625	x	x
0x63D0	0x6419	0 / +	0x6800	2048	x	x
0x7800	0x7800	0 / +	0x7C00	INFINITY	1	x
		1 / -	0x0000	ZERO	x	x
0x0401	0x0400	1 / -	0x0000	UNDERFLOW	x	1